

Review Quiz

Módulo 1: IV, GMM, Panel Data, Standard Errors & Bootstrap

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Question 1 — OLS Foundations

Which condition is central to the **consistency** of the OLS estimator, and whose violation is precisely what motivates instrumental variables?

- a) Normality of the error term
- b) Orthogonality: the regressors must be uncorrelated with the error term
- c) A sample size larger than 30
- d) The presence of an intercept in the model

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- d) The presence of an intercept in the model

Correct answer: b) Orthogonality between regressors and the error

Question 2 — Instrument Validity

An instrument z is considered **valid** when it satisfies which two properties?

- a) Linearity and homoscedasticity
- b) Normality and independence from the outcome variable
- c) Consistency and unbiasedness
- d) Exogeneity (uncorrelated with the structural error) and relevance (correlated with the endogenous regressor)

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Correct answer: d) Exogeneity and relevance

Question 3 — Exclusion Restriction

What does the **exclusion restriction** require of a valid instrument z in the structural equation for y ?

- a) That z affect y *only indirectly*, through its effect on x , with no direct effect on y
- b) That z be uncorrelated with the endogenous regressor x
- c) That z be measured without error
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Correct answer: a) z affects y only through x , never directly

Question 4 — Identification

A linear IV model is said to be **exactly identified** when...

- a) there are strictly more instruments than endogenous regressors
- b) no valid instruments are available
- c) the number of excluded instruments r equals the number of endogenous regressors q ($r = q$)
- d) the instruments are perfectly correlated with each other

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Correct answer: c) $r = q$

Question 5 — Weak Instruments

Theoretically, why does a **weak instrument** (low first-stage F -statistic) cause the 2SLS estimator to behave poorly in finite samples?

- a) Because weak instruments make OLS inconsistent
- b) Because the 2SLS bias approaches the OLS bias as instrument strength (and F) falls, per Staiger & Stock (1997)
- c) Because a low F -statistic implies the error term is non-normal
- d) Because weak instruments violate the exclusion restriction by definition

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Correct answer: b) 2SLS bias $\rightarrow$ OLS bias as instrument strength falls

Question 6 — GMM Foundations

The Generalized Method of Moments (GMM) framework is built on which foundational condition?

- a) The requirement that residuals be normally distributed
- b) The assumption that all regressors are exogenous by construction
- c) The condition that $N < T$
- d) The population moment condition $E[h(w_i, \theta_0)] = 0$

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Correct answer: d) $E[h(w_i, \theta_0)] = 0$

Question 7 — GMM: Optimal Weighting

Which choice of weighting matrix W_N minimizes the asymptotic variance of the GMM estimator, and what happens to this choice when the model is exactly identified?

- a) $W_N = S_0^{-1}$ is optimal in general, but when $r = q$ the choice of W_N becomes *irrelevant* and GMM coincides with simple IV
- b) $W_N = I$ is always optimal, regardless of identification
- c) The optimal W_N is a diagonal matrix of regressor variances
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Correct answer: a) $W_N = S_0^{-1}$ optimal; irrelevant when $r = q$

Question 8 — Overidentification (Hansen/Sargan) Test

What does the Hansen/Sargan overidentification (J) test evaluate, and what does a large test statistic suggest?

- a) Whether the sample size is adequate; a large value indicates insufficient data
- b) Whether the errors are homoscedastic; a large value indicates heteroscedasticity
- c) Whether the overidentifying moment conditions hold; a large value suggests invalid instruments or model misspecification
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Correct answer: c) Tests overidentifying restrictions; large value $\Rightarrow$ possible invalid instruments

Question 9 — Hausman (Wu–Hausman–Durbin) Test

In the Wu–Hausman test comparing OLS and 2SLS estimates, what is the **null hypothesis**?

- a) The instruments are weak
- b) The suspect regressor is exogenous, i.e. OLS is already consistent
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Correct answer: b) Regressor is exogenous (OLS consistent)

Question 10 — Panel Data: Random Effects

In the random-effects model, individual effects α_i and idiosyncratic errors ε_{it} combine into a composite error. What is the implied **equicorrelation** between the composite errors of the *same* individual across two different periods?

- a) Always zero
- b) $\frac{\sigma_{\varepsilon}^2}{\sigma_{\alpha}^2}$
- c) Always equal to one
- d) $\frac{\sigma_{\alpha}^2}{\sigma_{\alpha}^2 + \sigma_{\varepsilon}^2}$

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Correct answer: d) $\sigma_{\alpha}^2 / (\sigma_{\alpha}^2 + \sigma_{\varepsilon}^2)$

Question 11 — Choosing Fixed vs. Random Effects

Suppose the Hausman test **fails to reject** its null hypothesis (individual effects appear uncorrelated with the regressors). Which estimator should then be preferred, and why?

- a) Fixed effects, because it is always the safer choice regardless of the test result
- b) Pooled OLS, because the test confirms that no unobserved heterogeneity exists at all
- c) Random effects, because under the null it is consistent *and* more efficient than fixed effects, exploiting both within- and between-individual variation
- d) Neither estimator is valid; only first differences can be used

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- d) Neither estimator is valid; only first differences can be used

Correct answer: c) Random effects (consistent & more efficient under the null)

Question 12 — Fixed vs. Random Effects: Practical Use

Which statement best characterizes the practical trade-off between the fixed-effects and random-effects estimators?

- a) Fixed effects is valid only for balanced panels, while random effects works for any panel structure
- b) Fixed effects controls for *all* time-invariant unobserved heterogeneity (even unmeasured factors) but cannot estimate the effect of time-invariant regressors; random effects *can* estimate time-invariant regressors, but only under the stronger assumption that individual effects are uncorrelated with the regressors
- c) Random effects removes all endogeneity concerns automatically, so it should always be preferred over fixed effects
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Correct answer: b) FE: robust but no time-invariant coefficients; RE: needs uncorrelatedness, but keeps them

Question 13 — Robust & Clustered Standard Errors

The **Moulton factor**, which shows how much standard errors are underestimated when within-cluster correlation is ignored, is approximately given by which expression (with cluster size n and intraclass correlation ρ_e)?

- a) $1 - n\rho_e$
- b) n/ρ_e
- c) ρ_e^2
- d) $1 + (n - 1)\rho_e$

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Correct answer: d) $1 + (n - 1)\rho_e$

Question 14 — Bootstrap Methods

What is the core theoretical idea behind bootstrap methods, as referenced in the standard-errors material (Cameron & Trivedi, Ch. 11) for cases where analytical variance formulas are unreliable (e.g. too few clusters)?

- a) Repeatedly resampling *with replacement* from the observed data (or from whole clusters) to approximate the sampling distribution of an estimator directly from the data, rather than relying on an asymptotic formula
- b) Deleting outlier observations from the dataset before running OLS
- c) Estimating the model on a single random subsample and reporting that as the final result
- d) A method for choosing the optimal number of instruments in a GMM model

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Correct answer: a) Resample (with replacement) to approximate the sampling distribution

Answer Key

1. b
2. d
3. a
4. c
5. b
6. d
7. a
8. c

9. b
10. d
11. c
12. b
13. d
14. a